

dealer at a broking firm for, say, 5 lakh shares within a price range, it would be the dealer's responsibility to ensure that the order was executed in a way that the price and volume objectives were fulfilled. This meant having to execute the order in separate chunks by gauging the demand-supply situation from the screen, so as not to draw the market's attention.

With algo trading, all that would be taken care of by the software. FIIs and foreign brokerages clearly had the edge when it came to algo trading because they could afford to hire the best talent to develop proprietary software. DMA was slow to pick up initially, since the stock exchanges wanted to scrutinize the algorithms before they were allowed to run on their systems. The exchanges had a legitimate concern that algos without adequate safeguards could wreak havoc in the market. Having spent millions on their proprietary trading software, foreign players were reluctant to reveal the secret sauce of their algos. But once the stock exchanges relaxed this rule and the players did not have to reveal the specifics of their algos, DMA took off.

Domestic arbitrage firms, with their huge teams of mercenary day traders, never stood a chance in what was the equivalent of an arms race in the financial markets. For one, humans could never match the speed of programmed machines, which could spot arbitrage opportunities in a fraction of a second and execute them even before a day trader could blink. Local firms did make an attempt to fight back though, through commonly available, off-the-shelf trading software. The trouble with the generic algos was their lack of uniqueness, so that one user firm's algo trading differed little from another's. And they were no match for the high-end trading software developed by FIIs and foreign brokerages with their team of highly skilled software engineers.

There was another missing link on the road to superfast trading. Even the best of trading software would not be very effective unless the orders could be executed in the shortest possible time – measured in milliseconds – to beat competing algos. That hurdle was overcome when first NSE, and later BSE, began offering a facility known as 'co-location'. Under this, brokers could, for a fee, get to place their servers close to the exchange's trading engine. This would give the broker faster access to the buy-sell quotes streamed from the exchange and also send the executed trades back to the exchange quicker for confirmation. There was a slight lag – in milliseconds, not visible to the human eye – with which the data streamed from the